Akoizo Original Practice - 2027 Rotation Form 1 - Answer key

Division C | Anatomy And Physiology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Aspirated peanut most likely lodges in which bronchus and why?

Unit: Airways: From Nose to Bronchioles.
Worked-case data: Tube A has the baseline radius. Tube B has 80% of that radius. Both have the same length, viscosity, and driving pressure.

A. Right, wider and more vertical

B. Left, more horizontal

C. Trachea only

D. Left, narrower

Correct: A

2. (2 points) Vital capacity equals:

Unit: Ventilation, Volumes, and Spirometry.
Worked-case data: Person A breathes 500 mL twelve times each minute. Person B breathes 250 mL twenty-four times. Both have 150 mL anatomical dead space per breath.

A. ERV + RV

B. TV + RV

C. TV + IRV

D. IRV + TV + ERV

Correct: D

3. (2 points) Reversible obstruction with eosinophils best fits:

Unit: Obstructive Disease: Asthma, Bronchitis, Emphysema, COPD.
Worked-case data: Dataset A has variable wheeze, reduced expiratory ratio, and largely preserved diffusion capacity. Dataset B has persistent obstruction, increased residual volume, and reduced diffusion capacity.

A. Pneumothorax

B. Emphysema

C. Asthma

D. Fibrosis

Correct: C

4. (2 points) Pepsinogen is converted to pepsin by:

Unit: Alimentary Canal: Mouth to Stomach.
Worked-case data: An educational case describes loss of parietal-cell function with adequate dietary B12 but poor B12 absorption.
Separate investigation observations:
Diet record: The case provides adequate vitamin B12 intake.
Gastric finding: Parietal-cell function is greatly reduced.
Distal finding: The terminal ileum is anatomically present, but normal B12 uptake is impaired.

A. Bile

B. HCl/acid pH

C. Bicarbonate

D. Insulin

Correct: B

5. (2 points) Duodenal ulcer pain classically:

Unit: Ulcers, GI Bleeding, and GI Cancers.
Worked-case data: Case A supplies impaired mucus protection and a focal mucosal defect. Case B supplies an invasive colonic lesion that narrows the lumen.
Separate investigation observations:
Mucosal image description: A focal defect extends through the mucosa.
Protective function: Mucus-bicarbonate protection is reduced.
Lumen assessment: No obstructing tissue mass is described.

A. Better with food/antacid

B. No relation

C. Only nocturnal

D. Worse with food

Correct: A

6. (2 points) B cells mature in the ___; T cells mature in the ___.

Unit: Central Immune Organs and Lymph Flow.
Worked-case data: A small skin abrasion introduces material into tissue fluid of the left foot. Trace a plausible route toward immune surveillance and return to blood.
Separate investigation observations:
Origin: Antigen enters interstitial fluid around a skin abrasion.
Transport: Afferent lymph carries antigen-bearing cells from the tissue.
Cell encounter: Mature naive lymphocytes meet presented antigen in organized lymphoid tissue.

A. Spleen; node

B. Node; spleen

C. Thymus; marrow

D. Marrow; thymus

Correct: D

7. (2 points) MAC is formed by:

Unit: Innate Immunity and Complement.
Worked-case data: A hypothetical dataset shows poor complement-mediated coating of microbes, while antibody production is present. A second dataset shows coating but failure of terminal membrane attack.
Separate investigation observations:
Recognition assay: Microbial targets are recognized.
Coating assay: C3b deposition on targets is markedly reduced.
Phagocyte assay: Phagocytes work better when an external opsonin is supplied.

A. Histamine

B. C3b only

C. C5b-9

D. IgE alone

Correct: C

8. (2 points) AIDS is defined by CD4 under ___ or opportunistic illness.

Unit: Immunodeficiency: HIV, SCID, and CVID.
Worked-case data: A hypothetical case has circulating B cells but repeatedly poor antibody responses to supplied antigen challenges. T-cell measurements are also provided and appear preserved.
Separate investigation observations:
Cell inventory: B cells are present in the sample.
Challenge response: Antibody output after repeated specified challenges is poor.
Comparison: The supplied T-cell functional comparison is preserved.

A. 500

B. 350

C. 200

D. 50

Correct: C

9. (4 points) Which cells and structures drive mucociliary clearance?

Unit: Airways: From Nose to Bronchioles.
Worked-case data: Tube A has the baseline radius. Tube B has 80% of that radius. Both have the same length, viscosity, and driving pressure.

answer: Ciliated pseudostratified cells plus goblet mucus; cilia beat toward pharynx. (4 points).

10. (2 points) FEV1/FVC 0.55 with reduced FEV1 suggests:

Unit: Ventilation, Volumes, and Spirometry.
Worked-case data: Person A breathes 500 mL twelve times each minute. Person B breathes 250 mL twenty-four times. Both have 150 mL anatomical dead space per breath.

A. Normal

B. Mixed only

C. Restriction

D. Obstruction

Correct: D

11. (2 points) A teaching spirometry case gives tidal volume 520 mL, dead space 150 mL and respiratory rate 12/min. Find alveolar ventilation.

answer: 4440 mL/min (2 points).

Numerical target 4440; absolute tolerance 8.88. Accepted units: , mL/min.

12. (2 points) Use the same case to calculate minute ventilation: tidal volume 520 mL and rate 12/min.

answer: 6240 mL/min (2 points).

Numerical target 6240; absolute tolerance 12.48. Accepted units: , mL/min.

13. (2 points) A model narrows an airway while pressure difference and other conditions are fixed. Explain the likely airflow consequence.

answer: Narrowing increases resistance and generally reduces flow for a fixed driving pressure; real airways also depend on compliance and flow regime. (2 points).

14. (2 points) Describe how pleural fluid assists ordinary breathing.

answer: It lubricates pleural surfaces so they can slide with low friction while the pleural coupling helps chest and lung motion interact. (2 points).

15. (2 points) Why does the small intestine combine villi, microvilli and a thin absorptive epithelium?

answer: They increase effective exchange area and support absorption while limiting transport distance, with blood/lymph pathways carrying absorbed products. (2 points).

16. (2 points) Distinguish the digestive roles of bile and pancreatic lipase.

answer: Bile salts aid emulsification and micellar handling of lipids; lipase enzymatically hydrolyzes triglycerides. Bile is not itself a lipase. (2 points).

17. (2 points) Explain why a pancreatic bicarbonate secretion can assist digestion of acidic stomach contents.

answer: It helps neutralize acid in the small intestine and supports an appropriate environment for intestinal and pancreatic enzyme activity. (2 points).

18. (2 points) Compare innate and adaptive immune responses without calling either one universally faster or stronger in every situation.

answer: Innate mechanisms provide broad early recognition and defense; adaptive lymphocyte responses are antigen-specific and can generate memory, with timing affected by prior exposure. (2 points).

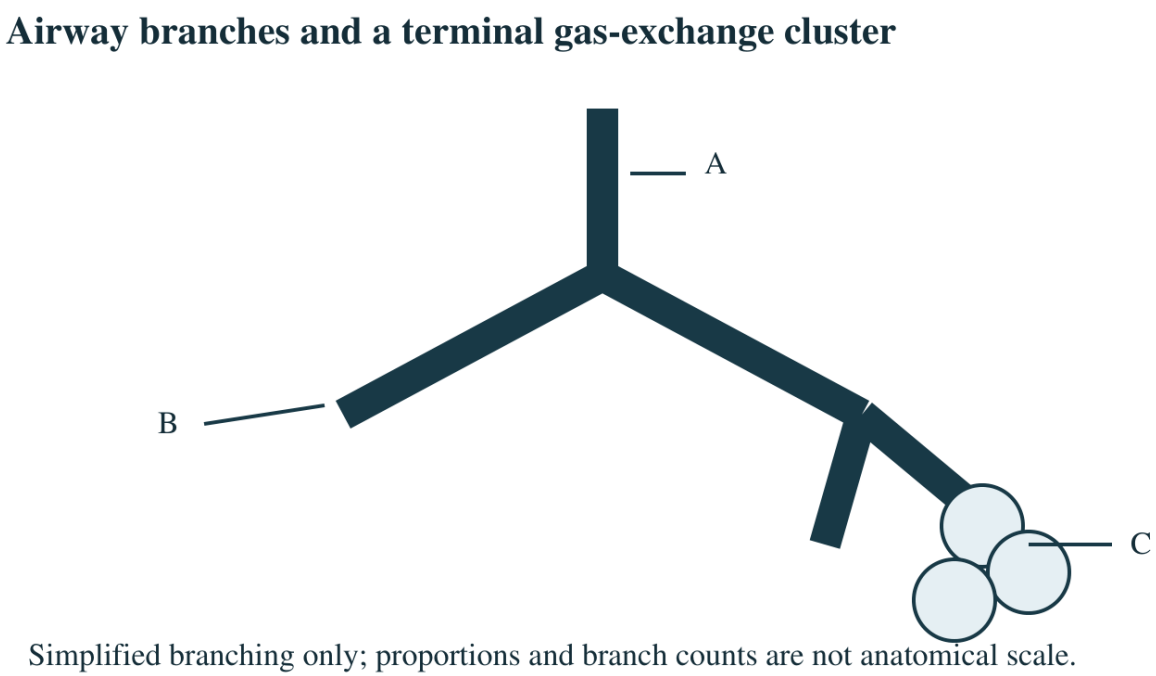
19. (2 points) Explain why a secondary antibody response can differ from a primary response.

answer: Memory B cells and other adaptive changes can produce a faster, larger and higher-affinity response after re-exposure to the same antigen. (2 points).

20. (2 points) In a teaching case, helper T-cell activation is impaired. Give one potential consequence for a coordinated adaptive response.

answer: Impaired helper signaling can reduce effective B-cell activation/class switching or other coordinated immune responses; the consequence depends on the pathway and context. (2 points).

21. (2 points) Use Figure F1. Name the structure labeled A.



F1. Respiratory branching schematic (original synthetic teaching figure)

answer: Trachea. (2 points).

22. (2 points) Use Figure F1. Which labeled region most directly provides the thin gas-exchange interface?

Refer to F1. Respiratory branching schematic (original synthetic teaching figure) shown with the first question in this figure group.

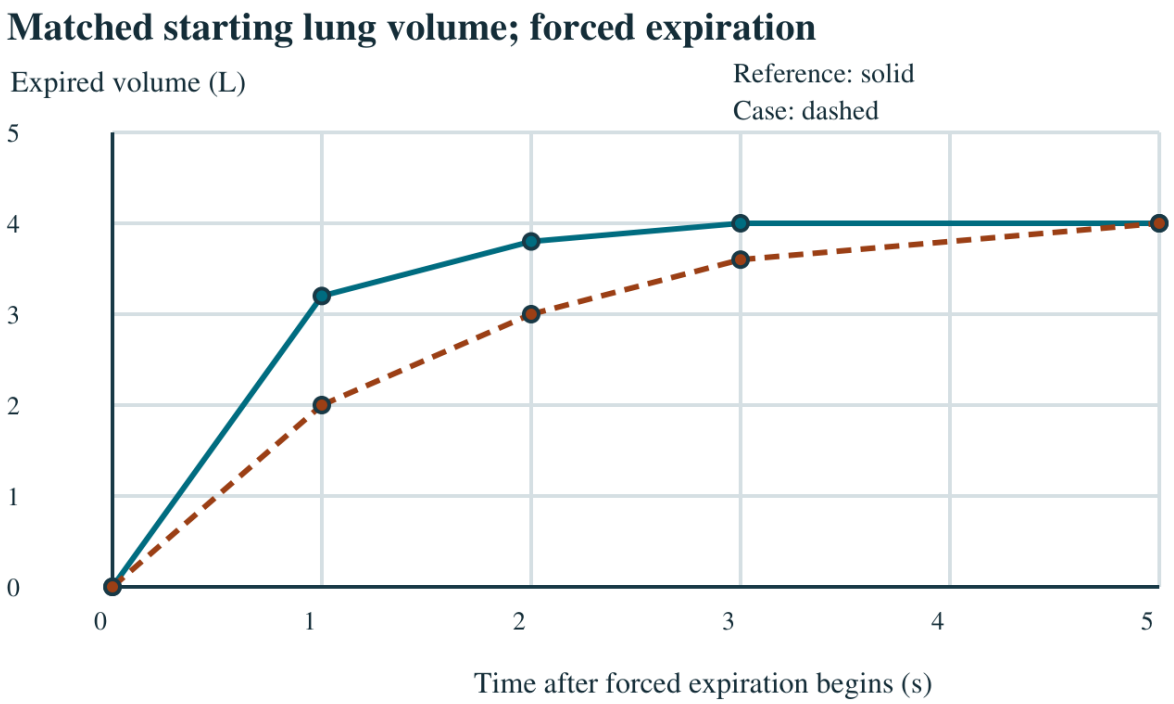
answer: C, the alveolar cluster. (2 points).

23. (2 points) Use Figure F1. Trace inspired air from A toward C using the branching arrangement.

Refer to F1. Respiratory branching schematic (original synthetic teaching figure) shown with the first question in this figure group.

answer: Air passes from trachea through a main bronchus and progressively smaller bronchi/bronchioles to alveoli; the drawing omits many intermediate branches. (2 points).

24. (2 points) Use Figure F2. Calculate case FEV1/FVC as a percentage.



F2. Forced expiratory volume curves (original synthetic teaching figure)

answer: 50 % (2 points).

Numerical target 50; absolute tolerance 0.15. Accepted units: , %.

25. (2 points) Use Figure F2. Which curve shows slower early emptying, and what pattern does that support studying?

Refer to F2. Forced expiratory volume curves (original synthetic teaching figure) shown with the first question in this figure group.

answer: The case/dashed curve; reduced early expired volume relative to final capacity supports studying an obstructive pattern, without establishing a specific disease by itself. (2 points).

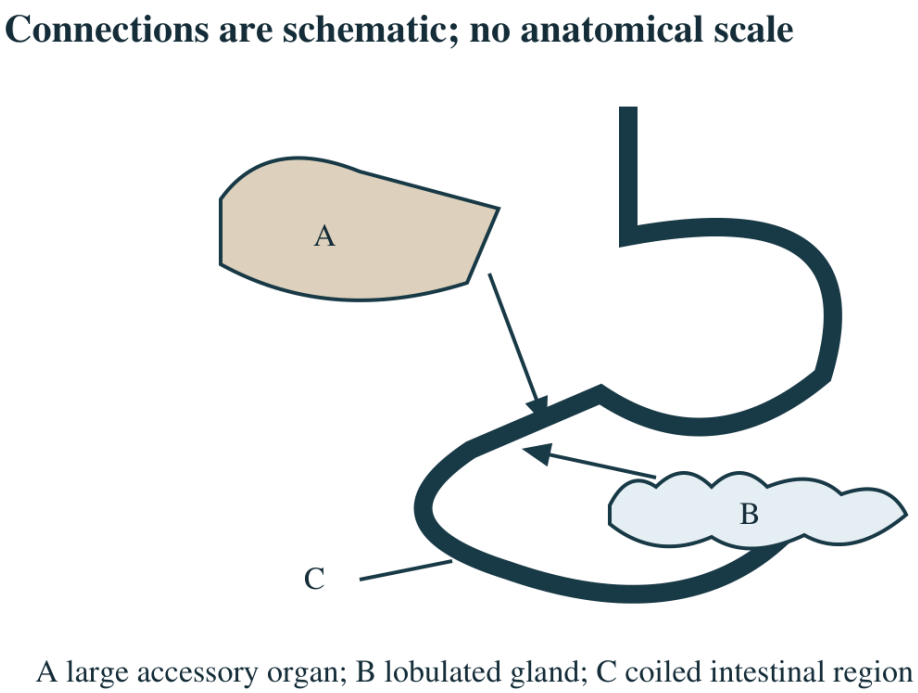
26. (2 points) Use Figure F2. Find reference-minus-case expired volume at1 s.

Refer to F2. Forced expiratory volume curves (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1.2 L (2 points).

Numerical target 1.2; absolute tolerance 0.0036. Accepted units: , L.

27. (2 points) Use Figure F3. Which labeled organ produces bile? Give its name.



F3. Digestive organ relationships (original synthetic teaching figure)

answer: A, the liver. (2 points).

28. (2 points) Use Figure F3. Name B and explain one digestive secretion it delivers toward the shown intestinal connection.

Refer to F3. Digestive organ relationships (original synthetic teaching figure) shown with the first question in this figure group.

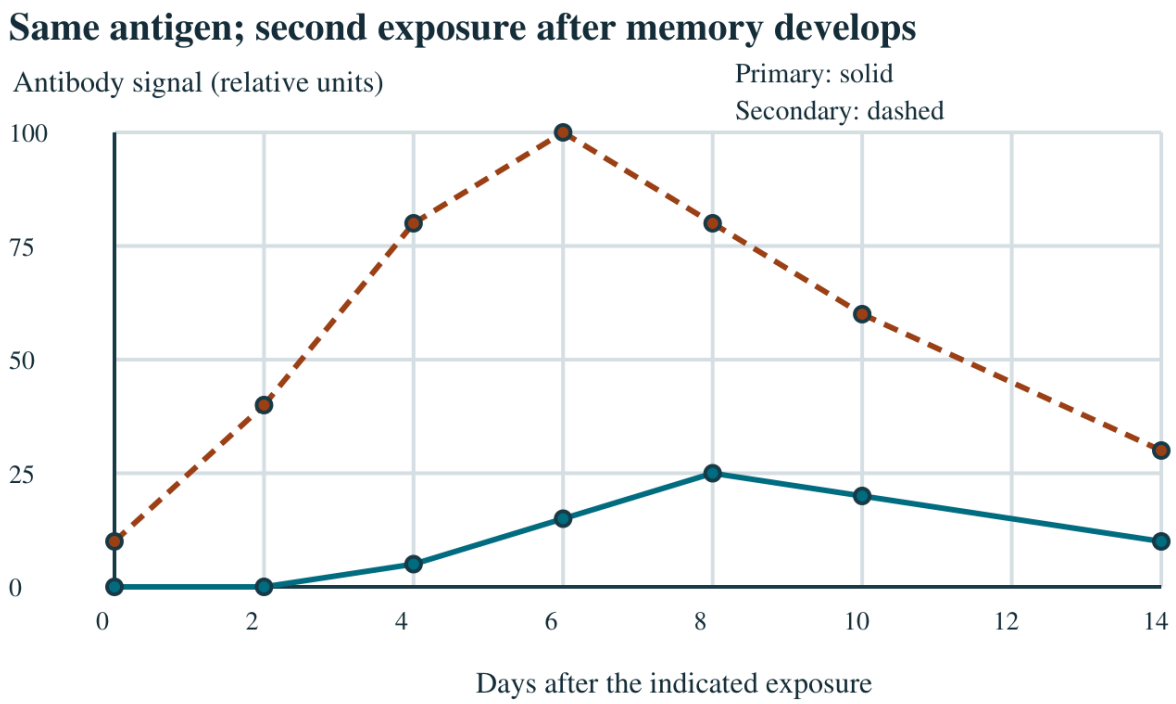
answer: Pancreas; digestive enzymes and bicarbonate enter the duodenum through pancreatic ducts. (2 points).

29. (2 points) Use Figure F3. Which labeled region is most directly associated with most nutrient absorption?

Refer to F3. Digestive organ relationships (original synthetic teaching figure) shown with the first question in this figure group.

answer: C, the small-intestinal region. (2 points).

30. (2 points) Use Figure F4. Read the ratio secondary-peak / primary-peak signal.



F4. Primary and secondary antibody responses (original synthetic teaching figure)

answer: 4 (2 points).

Numerical target 4; absolute tolerance 0.012. Accepted units: .

31. (2 points) Use Figure F4. At day4, calculate secondary-minus-primary signal.

Refer to F4. Primary and secondary antibody responses (original synthetic teaching figure) shown with the first question in this figure group.

answer: 75 relative units (2 points).

Numerical target 75; absolute tolerance 0.225. Accepted units: , relative units.

32. (2 points) Use Figure F4. Explain how the curve comparison is consistent with adaptive immune memory.

Refer to F4. Primary and secondary antibody responses (original synthetic teaching figure) shown with the first question in this figure group.

answer: The secondary response is faster and larger, consistent with memory lymphocyte activation; the schematic alone does not identify every cellular mechanism. (2 points).